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# 324454(25)

## B. E. (Fourth Semester) Examination, Nov.-Dec. 2016

**(New Scheme)**

**(Elect. Branch)**

## DIGITAL ELECTRONICS & LOGIC DESIGN

***Time Allowed : Three hours***

***Maximum Marks : 80***

***Minimum Pass Marks : 28***

***Note*** *: All questions are compulsory. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d) of each unit. All questions carry equal marks.*

### Unit - I

1. (a) Define self complementing code and name one self complementing code. 2

(b) A message signal is received as 1000011. Check for any error and if there is any error then correct it using even parity. 7



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(c) Do as directed : 7

(i) $(142 \cdot 96)_{10} \longrightarrow (\quad)_8$

(ii) $(1100011010)_{Gray} \longrightarrow (\quad)_2$

(iii) Represent CSVTU in ASCII code.

(iv) Represent 459 in 2421 code.

(v) Find 10's complement of $(935)_{11}$.

(vi) $(2DE \cdot CA6)_{16} \longrightarrow (\quad)_8$

(vii) Represent 368 in 84-2-1 code.

(d) (i) Reduce $BC + AC' + AB + BCD$ to 4 literals.

(ii) Implement $XY + X'Y' + Y'Z$ only with AND and NOT gates.

(iii) Implement Ex-OR with minimum number of NAND gates.

(iv) State De Morgan's theorem. 7

### Unit - II

2. (a) Prove that dual of Ex-OR gate is equal to it's complement. 2

(b) Find minimum expression in SOP for :



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$$F(A, B, C, D) = \Sigma(3, 4, 5, 7, 9, 13, 14, 15)$$

and implement expression (simplified) only with NAND gates. 7

(c) Simplify using Tabulation method : 7

$$F(A, B, C, D) = \Sigma(0, 1, 5, 8, 11) + d(4, 12, 14)$$

(d) Find minimum expression in POS for : 7

(i) $F_1 = f + g$

(ii) $F_2 = f \cdot g$

where $f = w\,xy' + y'z + w'\,yz' + x'yz'$ and

$$g = (w + x + y' + z')\,(x' + y' + z)\,(w' + y + z')$$

## Unit - III

3. (a) Define Combinational Logic Circuit. 2

(b) Implement a 5 × 32 decoder circuit using 3 × 8 and 2 × 4 decoders. 7

(c) Design a 4 bit Binary to Gray code converter circuit. 7

(d) Implement a Full adder circuit using two half address. 7



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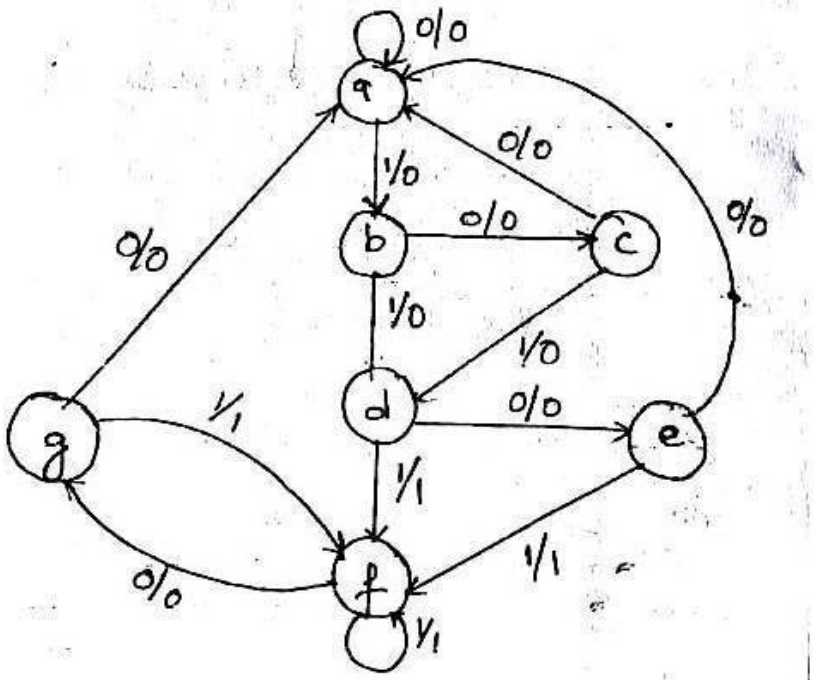
## Unit - IV

4. (a) What does J, K stands for in JK flip-flop? 2

(b) Design a MOD-12 counter (up) circuit using T flip-flop. 7

(c) Explain SIPO register configuration and it's working. 7

(d) Reduce the following state diagram to minimum states : 7



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## Unit - V

5. (a) Define Figure of Merit. 2

(b) Draw TTL circuit with totem pole output and explain it's working. 7

(c) Explain CMOS-NAND gate? 7

(d) Draw neat diagram of ECL and explain it briefly. 7

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